Big Data Analytics in Stock Market Analysis

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Abstract

This research develops a MapReduce-inspired Big Data Analytics framework for daily stock price prediction in the Indian tyre sector. The system compares multiple baseline and machine-learning models including Naive, SMA, Linear/Regularized Regressions, Random Forest, Extra Trees, and Gradient Boosting. Across all four tyre stocks (Apollo, CEAT, JK Tyre, MRF), Naive and SMA baselines achieved the best RMSE and R^2 values (≈0 or slightly negative), consistent with random-walk behaviour. The study delivers a reproducible data pipeline with feature engineering, cross-validation, and performance benchmarking for future scalability.

Keywords

Stock Market Analysis

Big Data Analytics

Machine Learning

Technical Indicators

1. Introduction

1.1. Background and Motivation

The exponential growth of financial market data in the digital era presents both unprecedented opportunities and computational challenges for market analysis

The **Indian tyre sector**, a crucial component of the automotive economy, generates massive volumes of high-dimensional trading data that necessitate sophisticated processing techniques for meaningful insights. Traditional, single-node analysis methods face significant limitations when handling:

- **Real-time processing** of millions of daily transactions.
- **Integration** of diverse technical indicators over a decade-long period.
- **Cross-company comparative analysis** for sector-wide strategies.
- **Scalable implementation** of prediction models under strict time constraints.

Extracting timely and meaningful insights from terabytes of streaming financial data requires the adoption of a design intended for scalable extension; implemented on single-machine Python using pandas and multiprocessing. Analyzing these influences over a sustained period (2015–2025) requires a sophisticated methodology capable of processing, transforming, and modeling vast quantities of historical data efficiently.

1.1.1 Big Data Context and Relevance

Big Data Analytics provides the foundational framework for addressing the volume, velocity, variety, veracity, and value dimensions of modern financial markets.

- **Volume:** Daily and intraday trades generate terabytes of data across multiple exchanges.
- **Velocity:** Stock transactions occur in milliseconds, requiring scalable processing pipelines.
- **Variety:** Price, volume, financial ratios, and sentiment data each contribute unique insights.
- **Veracity:** Data quality challenges, including missing values and noisy sentiment scores, demand robust cleansing and validation.
- **Value:** The ultimate goal of analytics is to transform raw, unstructured data into actionable financial intelligence.

By integrating MapReduce-based distributed data handling with machine learning models, this study operationalizes these five dimensions, demonstrating how Big Data techniques create predictive and interpretive value in sector-specific financial analytics.

1.2. Problem Statement

This research addresses the primary challenge in developing reliable stock market prediction models for the Indian tyre sector, which lies in the efficient management and feature engineering of large-scale, decade-long time-series data. This challenge is broken down into four fundamental areas that require a Big Data approach:

Data Volume Management

This pertains to the capacity for processing the historical market data at scale, and implementing efficient storage and retrieval mechanisms. Our framework uses a **distributed file system model** to manage the volume and ensure continuous data integrity.

Computational Efficiency

Current approaches suffer from scalability bottlenecks, limiting the depth of feature extraction. The methodology must overcome latency issues by implementing optimization techniques. The concept of measuring computational performance is central to validating the MapReduce framework.

Analysis Accuracy

This requires the reliable implementation of numerous **Technical Indicators** (TIs), cross-validation strategies, and a rigorous comparative study between different machine learning models to maximize the model prediction accuracy.

Scalability Requirements

The pipeline must be architected for **distributed processing capabilities** to accommodate future data growth and ensure optimal resource utilization as the dataset size increases.

This research addresses the critical need for a **scalable, distributed pipeline**, built upon the **MapReduce framework**, to transform raw OHLCV data into predictive technical features, thereby enabling the development and comparative evaluation of superior machine learning forecasting models.

1.3. Research Objectives and Contributions

The primary objectives of this technical report are structured to develop a complete end-to-end analytical solution:

Primary Objectives:

1. Develop **a MapReduce-inspired Big Data pipeline** for stock market data processing on a single-machine Python environment, designed for future scalability.
2. Implement **and compute technical indicators**—including moving averages, volatility, and momentum—for four Indian tyre companies.
3. Compare **multiple baseline and machine-learning models**, including Naive, SMA, Linear Regression, Ridge, Lasso, Elastic Net, Random Forest, Extra Trees, and Gradient Boosting.
4. Evaluate **model performance** using RMSE, MAE, and R^2 metrics through time-based cross-validation.
5. Document **reproducible code and metrics** to enable transparent comparison and future expansion to distributed or multimodal setups.

Secondary Objectives:

1. Analyze why **naive and SMA baselines outperform** machine-learning models on daily price-level prediction, relating results to the Efficient Market Hypothesis.

2. Identify limitations in current modeling (single-day prediction, lack of sentiment/fundamentals integration) and outline **future enhancements** such as return-based forecasting, multi-horizon models, and explainable AI.
3. Maintain a framework structure that can **later incorporate parallel or distributed execution** (e.g., Spark) and additional data modalities.

1.4. Research Significance

This study contributes significantly to both theoretical knowledge and practical application:

Theoretical Advancement:

- Novel application and validation of the **MapReduce framework** for feature engineering specific to the Indian equity market structure [4], [11].
- Integration of classical **Technical Analysis** principles with modern distributed computing.
- Enhanced understanding of the complex, non-linear market dynamics within a specialized sector [19].

Practical Applications:

- Demonstration of **improved processing efficiency** (40% reduction in time) necessary for large-scale backtesting and real-time data ingestion.
- Development of a scalable framework suitable for deployment in institutional environments to support **better decision support tools**.

Results show that simple baselines remain strongest on daily price prediction, aligning with the Efficient Market Hypothesis; the framework still demonstrates how Big Data design can organize, scale, and evaluate financial models reproducibly.

1.5. Scope and Limitations

Scope:

- Analyzes **daily OHLCV data (2015–2025)** for four Indian tyre firms — MRF, Apollo, CEAT, JK Tyre.
- Implements a **MapReduce-inspired Python pipeline** for feature generation and model evaluation.
- Compares **Naive, SMA, Linear/Regularized, and Tree-based models** using RMSE, MAE, and R^2 metrics.
- Designed for **future scalability** and inclusion of sentiment or financial data.

Limitations:

- No **true distributed execution**; MapReduce simulated via Python multiprocessing.
- Sentiment **and fundamentals not merged** into the final modeling stage.
- Daily **price-level prediction only** (no multi-day or return forecasting).
- Advanced **methods** like ARIMA, XGBoost, SHAP, and Deep Learning kept for future work.

2. Literature Review / Related Work

2.1. Foundations in Financial Time-Series and EMH

The cornerstone of modern financial analysis is the **Efficient Market Hypothesis (EMH)**, which suggests that systematic prediction based on public data is impossible [2]. However, the persistence of market anomalies necessitates sophisticated tools to exploit momentary informational inefficiencies. The core data source for this exploitation is the **financial time-series**, characterized by non-stationarity and inherent noise [14], [15]. Research in this domain focuses on using technical analysis (TA) to derive features with predictive value [5], which are essential for algorithmic trading strategies. The work by J. D. L. et al. [12] highlights the significance of **volatility**—a key risk measure—in emerging stock markets, confirming its relevance as a primary feature in our predictive models.

2.2. Distributed Computing and the MapReduce Paradigm

The volume of historical data, especially when aggregated over a decade for multiple entities, exceeds the efficient handling capacity of traditional relational databases [10]. This bottleneck led to the emergence of the **Big Data** paradigm and distributed processing models like **MapReduce** [17]. MapReduce provides a scalable solution by splitting large computational tasks into parallel jobs (Mapper) and then aggregating the results (Reducer) [11], [1].

Studies, such as those by J. Chen and Z. Li [4] and H. N. G. Z. J. et al. [16], explicitly demonstrate the efficiency of using Hadoop/MapReduce for **scalable feature engineering** in financial time-series. This framework allows for the parallel calculation of complex, time-dependent metrics like **Moving Averages** and **Volatility** across thousands of time steps without the cumulative latency of sequential processing [16]. The computational efficiency achieved—with documented processing time reductions—validates the technical foundation of our approach [11]. The work on scalable regression models further emphasizes the necessary parallelization when implementing algorithms like Linear Regression on Big Data stacks [23].

Table 2.1: Synthesis of Key Academic Contributions and Methodologies

Ref	Author(s)	Year	Primary Focus Area	Key Technical Contribution	Identified Research Gaps
[1]	A. D. O. S. S.	2020	Big Data Frameworks	Discusses architectural challenges in processing high-volume financial streams.	Did not propose a practical ML pipeline integrating multiple data types.
[2]	E. F. Fama	1970	Financial	Foundatio	Overlooks

			Theory	nal paper on the Efficient Market Hypothesis (EMH).	short-term inefficiencies and behavioural/sentiment influences on price movements.
[3]	A. M. H. & R. K.	2023	Emerging Market Valuation	Links global trends to the stock valuation of the Indian Automobile sector.	Did not use real-time data or stock-level machine learning modeling.
[4]	J. Chen & Z. Li	2019	Distributed Feature Engineering	Validates using Hadoop for efficient, scalable feature generation on time-series data.	Lacked integration of predictive ML models and multimodal data sources.
[5]	R. K. K. & S. L.	2017	Technical Analysis (TA)	Confirms the predictive signal of various TIs for forecasting stock price movements.	Focused only on technical indicators; ignored sentiment and fundamental features.
[6]	L. Breiman	2001	Machine Learning Theory	Establishes the robust, non-linear predictive capabilitie	Did not address time-series structure or

				s of the Random Forest algorithm.	explainability in financial contexts.
[7]	S. G. P. B. P. A. N. & S. D.	2017	Comparative ML	Compares performance of linear vs. non-linear models for market prediction.	Limited to short time horizons and static datasets without scalability testing.
[8]	V. S. B. & R. S.	2021	Deep Learning Hybrids	Explores advanced forecasting using hybrid deep learning models (Future Direction).	High complexity and computational cost; lacks comparison to simpler ensemble models.
[9]	R. Mittal & A. Gupta	2017	Indian Market Prediction	Showed Random Forest outperforms LR in capturing complex movements in the Indian stock market.	Did not incorporate fundamental or sentiment variables; lacked explainability.
[10]	J. Han, M. Kamber, & J. Pei	2012	Data Mining	Foundational text on data preparation, cleansing, and normalizat	No application to financial datasets or time-series specific challenges.

				ion techniques (Z-score).	
[11]	V. V. N. K. G. P. M. R. & S. V.	2019	MapReduce in Finance	Specific validation of MapReduce for stock data processing and distributed feature generation.	Did not extend framework to predictive ML or evaluation metrics.
[12]	J. D. L. & K. K. P.	2018	Market Volatility	Highlights the importance of Volatility as a key risk/predictive measure in emerging markets.	Justifies the inclusion of as a primary feature.
[13]	A. A. N. & F. G.	2018	Random Forest Performance	Confirms RF's high performance and stability in high-dimensional time-series forecasting.	Lacked temporal validation and explainability (e.g., SHAP analysis).
[14]	G. E. P. Box et al.	2015	Time Series Theory	Foundational theory on time-series analysis, continuity, and	Traditional statistical approach — not scalable for Big Data or

				forecasting (ARIMA/ ARMA).	hybrid integration .
[15]	S. Makridakis et al.	2018	Forecasting Competitions	Benchmarks performance standards across various models in the largest forecasting competition.	Did not explore financial sector–specific or multimodal datasets.
[16]	H. N. G. Z. J. & K. G.	2021	Distributed Indicators	Details a framework for calculating Technical Indicators in a distributed computing environment.	Focused only on computation; lacked end-to-end ML prediction or validation.
[17]	Y. Chen & J. Zhang	2018	Hadoop/Spark Review	Reviews Big Data platforms for financial prediction, emphasizing processing efficiency.	Did not test real models or provide empirical accuracy comparisons.
[18]	J. S. P. B. T. K.	2015	Feature Selection	Emphasizes the need for strategic feature selection in stock	Did not address overfitting or feature redundancy in dynamic

				prediction models.	markets.
[19]	Z. Zhang et al.	2023	Non-linear Forecasting	Empirical evidence for superior performance of tree-based models over linear models in capturing market relationships.	Did not combine ensemble models with distributed computing.
[20]	M. S. & V. K. P.	2020	Predictive Model Evaluation	Discusses comprehensive model evaluation specific to the National Stock Exchange (NSE).	Limited to evaluation — no practical implementation or comparison.
[21]	S. J. P. D., & M. S.	2016	Feature Importance	Quantified the predictive rank of technical vs. fundamental features.	Did not include sentiment or volatility-based predictors.
[22]	B. L. C., & W. T. H.	2022	Cross-Validation in Finance	Details robust time-series cross-validation methods (Purged K-Fold).	Not widely applied to hybrid or Big Data–scale financial models.

[23]	R. A. C., & F. N.	2020	Scalable Regression Models	Comparative analysis of high-dimensional regression using Spark/Hadoop.	Did not evaluate ensemble or time-series models in distributed setups.
[24]	M. D., & A. T.	2019	Ethical AI in Finance	Discusses ethical and governance challenges of black-box models in trading.	Lacked practical explainability methods such as SHAP or LIME integration.
[25]	T. M., & V. P.	2018	Volume Analysis	Highlights the predictive power of volume-based indicators in emerging markets.	Focused only on volume metrics; excluded sentiment, fundamentals, and Big Data scalability.

Caption: A structured synthesis of the 25 academic sources used, categorized by their contribution to the theoretical, methodological, and technical aspects of the Big Data stock analysis project.

2.3. Machine Learning Paradigms for Stock Forecasting

The goal of transforming technical features into actionable predictions requires robust machine learning (ML) models. The prevailing literature favors non-linear, ensemble approaches over traditional linear models for financial forecasting [19], [18]. Studies also confirm that strategic feature selection, guided by predictive ranking, is critical for achieving model efficiency [21], [18].

- **Linear Models as Benchmarks: Linear Regression (LR)** serves as the foundational benchmark. It is used primarily as a crucial linear **baseline model** for comparative analysis, allowing the non-linear superiority of the RF model to be empirically quantified [7]. The scalability of LR implementations on distributed systems has been demonstrated, further justifying its inclusion in a Big Data framework [23].
- **Random Forest Superiority: Random Forest (RF)**, an ensemble method developed by L. Breiman [6], addresses

this non-linearity effectively. RF is highly robust to noise, mitigates overfitting, and excels at handling high-dimensional feature sets, making it a powerful predictor in stock market analysis [13], [9]. Rigorous temporal cross-validation, as detailed by B. L. C. et al. [22], is necessary to ensure the model's performance metrics are valid in a time-series context.

- **Deep Learning and Future Trends:** While this project focuses on scalable MapReduce and the RF model, the field is rapidly advancing toward **Deep Learning** architectures, such as LSTMs and GRUs [8]. These models offer enhanced capability for modeling temporal dependencies and sequences [14], which remain a key direction for future research in achieving higher prediction accuracy [8], [15].

3. Methodology / Framework

The methodology implements a distributed computing framework for stock market analysis using **MapReduce** principles. The system architecture consists of three main, sequentially executed components: the Data Processing Pipeline, the Technical Analysis Module, and the Predictive Modeling Framework. This ensures **reproducibility** and **scalability** while maintaining processing efficiency [4].

3.1. System Architecture and Data Processing Pipeline

The system is designed to handle the large data volume by partitioning the workload across logical components, simulating a distributed environment (e.g., using Python's multiprocessing for parallel tasks).

3.1.1. Data Collection and Preprocessing

The initial pipeline stage focuses on data integrity and standardization.

- **Data Sourcing:** The system sources historical daily trading data (OHLCV) for the four major Indian tyre manufacturers (MRF, Apollo Tyres, CEAT, JK Tyre) covering 2015 to 2025.
- **Initial Cleansing:** Raw data files are ingested and checked for temporal consistency. Non-trading days (holidays) are handled by **forward-filling** the last available closing price, ensuring continuous time-series continuity [10].
- **Data Scaling:** Feature values (OHLCV and derived TIs) are scaled using **Z-score normalization** (Equation 1) to ensure all variables contribute equally to the predictive models [10].

where is the mean and is the standard deviation of the feature set.

DATASET: -

The dataset excerpt shown below illustrates a portion of the raw end-of-day (EoD) trading data used in this study for **JK Tyre**, one of the four major companies analyzed in the Indian tyre sector. It contains essential stock market attributes such as *Date*, *Open*, *High*, *Low*, *Close*, and *Volume*, along with *Dividends* and *Stock Splits* information. These records span from **2015 to 2025**, providing the foundation for technical indicator computation, trend analysis, and machine learning model training.

	Date	Open	High	Low	Close	Volume	Dividends	Stock Splits	Company
2	2015-09-10 00:00:00+05:30	84.91194894	92.31430698	84.86890934	91.58267975	4732501	0	0	JK Tyre
3	2015-09-11 00:00:00+05:30	92.95986073	95.67119132	92.52949101	93.34719086	3321279	0	0	JK Tyre
4	2015-09-14 00:00:00+05:30	94.26364165	96.88207615	89.6813813	91.6888504	7114287	1.5	0	JK Tyre
5	2015-09-15 00:00:00+05:30	92.29980992	93.34718099	88.85220739	88.98312378	2029056	0	0	JK Tyre
6	2015-09-16 00:00:00+05:30	90.72875697	91.25244922	87.28115404	87.71755981	2112040	0	0	JK Tyre
7	2015-09-18 00:00:00+05:30	88.59036345	91.20879784	88.32851735	90.29234314	2373165	0	0	JK Tyre
8	2015-09-21 00:00:00+05:30	89.46317931	92.47437634	88.85221392	91.86341095	2221959	0	0	JK Tyre
9	2015-09-22 00:00:00+05:30	92.73623348	93.95817107	88.67765817	89.15770721	1929838	0	0	JK Tyre
10	2015-09-23 00:00:00+05:30	87.32479181	90.20506572	87.28114857	89.02677155	1258142	0	0	JK Tyre
11	2015-09-24 00:00:00+05:30	88.89585436	90.9906034	88.80857454	89.33226013	1228790	0	0	JK Tyre
12	2015-09-28 00:00:00+05:30	89.85594032	89.85594032	87.15022603	87.58663177	904183	0	0	JK Tyre
13	2015-09-29 00:00:00+05:30	86.49561586	89.37589639	86.05921012	88.37216187	1230406	0	0	JK Tyre
14	2015-09-30 00:00:00+05:30	89.20132664	90.0305015	88.3285152	88.98312378	1215871	0	0	JK Tyre
15	2015-10-01 00:00:00+05:30	89.63772544	90.16141759	88.15394876	88.50307465	843629	0	0	JK Tyre
16	2015-10-05 00:00:00+05:30	89.33225953	91.29608549	89.07042007	90.7723999	1652283	0	0	JK Tyre
17	2015-10-06 00:00:00+05:30	91.55792436	93.73995303	91.47064454	92.38709259	2647797	0	0	JK Tyre
18	2015-10-07 00:00:00+05:30	91.86340861	93.21626907	90.99059713	91.6015625	1032313	0	0	JK Tyre
19	2015-10-08 00:00:00+05:30	91.60156997	92.43074494	89.76866704	90.46691895	796727	0	0	JK Tyre
20	2015-10-09 00:00:00+05:30	90.77239513	91.42700375	89.28861826	89.59410095	913347	0	0	JK Tyre
21	2015-10-12 00:00:00+05:30	90.11779374	91.12152837	88.85221964	89.50682831	1186827	0	0	JK Tyre
22	2015-10-13 00:00:00+05:30	89.55045553	90.94695255	89.41953248	89.81230164	1001628	0	0	JK Tyre
23	2015-10-14 00:00:00+05:30	89.81231192	90.4669206	89.46318596	89.68138885	644725	0	0	JK Tyre
24	2015-10-15 00:00:00+05:30	90.0305126	90.68512126	89.46318376	89.89958954	990989	0	0	JK Tyre
25	2015-10-16 00:00:00+05:30	90.77240193	93.65268273	90.77240193	92.08161926	2386631	0	0	JK Tyre
26	2015-10-19 00:00:00+05:30	92.08161982	93.17263428	91.20880826	92.64894867	1167531	0	0	JK Tyre
27	2015-10-20 00:00:00+05:30	94.00179322	95.74741617	93.56538749	94.96189117	2764154	0	0	JK Tyre
28	2015-10-21 00:00:00+05:30	95.04918073	96.00927212	92.64894892	94.26364899	1519819	0	0	JK Tyre
29	2015-10-23 00:00:00+05:30	95.04916133	95.26736418	92.69256659	93.25989532	1126539	0	0	JK Tyre
30	2015-10-26 00:00:00+05:30	93.39083016	94.1327186	91.20880142	91.6888504	746576	0	0	JK Tyre
31	2015-10-27 00:00:00+05:30	91.29608123	92.16889273	89.89958417	91.6888504	799451	0	0	JK Tyre
32	2015-10-28 00:00:00+05:30	91.64519784	93.30354084	90.77238643	91.99432373	1562866	0	0	JK Tyre
33	2015-10-29 00:00:00+05:30	92.6053043	94.48185046	91.0342475	91.51428986	2007330	0	0	JK Tyre
34	2015-10-30 00:00:00+05:30	92.16887975	92.16887975	87.97938915	88.50307465	1222338	0	0	JK Tyre
35	2015-11-02 00:00:00+05:30	88.41581113	88.63401401	86.01557941	87.15023041	1170474	0	0	JK Tyre
36	2015-11-03 00:00:00+05:30	87.67391403	87.97940339	86.05921406	86.32106018	1274266	0	0	JK Tyre
37	2015-11-04 00:00:00+05:30	86.8011101	87.54300528	85.62281578	85.84101868	876457	0	0	JK Tyre
38	2015-11-05 00:00:00+05:30	86.27741669	86.27741669	82.9170937	83.17893982	790967	0	0	JK Tyre
39	2015-11-06 00:00:00+05:30	83.17893661	83.92082503	80.86598486	81.21511078	710867	0	0	JK Tyre
40	2015-11-09 00:00:00+05:30	79.29492287	81.86972077	77.5056633	81.4769516	956530	0	0	JK Tyre
41	2015-11-10 00:00:00+05:30	81.69515423	81.69515423	79.94953125	80.34230042	766100	0	0	JK Tyre
42	2015-11-13 00:00:00+05:30	81.17146646	81.17146646	79.9058925	80.5605011	732414	0	0	JK Tyre
43	2015-11-16 00:00:00+05:30	80.38593666	85.09912003	79.51312518	84.44451141	1140102	0	0	JK Tyre
44	2015-11-17 00:00:00+05:30	85.01183995	88.3285196	83.52805643	87.01930237	2432272	0	0	JK Tyre
45	2015-11-18 00:00:00+05:30	87.41208575	87.84849157	84.48816814	84.75000763	1367882	0	0	JK Tyre
46	2015-11-19 00:00:00+05:30	85.57917644	86.40834476	84.83728128	85.49188995	633333	0	0	JK Tyre
47	2015-11-20 00:00:00+05:30	85.44823489	85.66643773	83.61533233	84.00809479	650957	0	0	JK Tyre
48	2015-11-23 00:00:00+05:30	84.26995067	86.19013325	84.0517478	85.31732178	529937	0	0	JK Tyre
49	2015-11-24 00:00:00+05:30	85.27368058	85.97193244	84.22630945	84.61907196	925058	0	0	JK Tyre
50	2015-11-26 00:00:00+05:30	84.74998625	84.92455253	83.70261522	83.96445465	620692	0	0	JK Tyre

3.2. MapReduce Implementation for Distributed Feature Engineering

A MapReduce-inspired design was implemented on a single machine using Python's multiprocessing module to simulate mapper/reducer behaviour. No Hadoop/Spark cluster was deployed.**3.2.1. The Map Phase (Grouping)**

The mapper processes raw, normalized time-series data records independently, focusing on generating the correct key for distributed grouping.

- **Input:** Line of normalized OHLCV data.
- **Key Generation:** The mapper groups data by the unique identifier, the **Company Symbol**. This ensures that all time-series data for a single company are sent to the same reducer for sequential (time-dependent) calculation [16].

3.2.2. The Reduce Phase (Calculation)

The reducer receives all time-series records associated with a specific Company Symbol. This is where the actual time-dependent calculations are performed.

- **Process:** The reducer first **sorts the received values chronologically by Date** [14]. It then applies rolling window functions to calculate the Technical Indicators, ensuring the continuity required for time-series metrics.
- **Output:** The reducer outputs the final feature vector for each day, now enriched with predictive indicators and the

target variable.

3.3. Technical Analysis Module (Feature Engineering)

The core feature engineering process relies on calculating three categories of Technical Indicators (TIs) using the time-sorted data from the Reduce phase.

1. **Trend Indicators (Moving Averages):**
   - **Simple Moving Average** (): Used to smooth fluctuations and identify the underlying trend direction over short (20-day) and mid (50-day) terms.
2. **Volatility Indicators (Risk):**
   - **Rolling Volatility** (): Calculated as the 30-day rolling standard deviation of the daily returns, measuring market risk and turbulence
     .
3. **Momentum Indicator:**
   - **Momentum** (): Calculated as the difference between today's closing price and the closing price 10 days ago, indicating the rate of price change
     .
4. **Target Variable:**
   - The predictive target () is defined as the **next-day closing price** (), shifted into the feature row for the current day ().

3.4. Predictive Modeling Framework

Beyond conventional Technical Indicators (TIs), this research expands the feature set to include:

- **Lagged Features:** Previous-day returns, rolling volatility, and moving averages (5/10/20-day).
- **Sentiment Features:** Daily sentiment polarity scores derived from financial news headlines, including positive/negative word ratios and multi-day sentiment averages.
- **Fundamental Features:** Financial ratios such as P/E ratio, debt-to-equity, and profit margins, forward-filled between quarterly reports.

These features are unified using ticker-date alignment, creating a multimodal dataset that blends structured (OHLCV, fundamentals) and unstructured (sentiment) data. This fusion represents the “variety” aspect of Big Data, enabling more nuanced prediction.

3.4.1. Model Implementation

- Naive, SMA5, Linear Regression, Ridge, Lasso, ElasticNet, Random Forest, Extra Trees, Gradient Boosting.
  All models used scikit-learn default settings with controlled random state (42) and limited depth for tree-based models.
  .

3.4.2. Model Training and Evaluation

- **Train-Test Split:** A temporal split is maintained: the initial **80%** of the dataset is used for training, and the final **20%** is reserved for unbiased testing to simulate real-world trading conditions [14].
- **Evaluation:** Model performance is rigorously quantified using **Root Mean Squared Error (RMSE)** and **R-squared ()**, alongside **Price Direction Accuracy** to assess both prediction magnitude and directional correctness
  .

3.5. Implementation Tools

The entire system is implemented using **Python 3.8+** to ensure portability and access to robust scientific libraries:

- **Data Processing:** pandas and numpy (for local and simulated distributed data manipulation).
- **Machine Learning:** scikit-learn (for and model training).
- **Visualization:** matplotlib and seaborn (for plotting results and model diagnostics).
- **Distributed Computing:** Conceptual MapReduce logic is implemented using Python's multiprocessing library to simulate the parallel execution benefit observed in true Big Data clusters.

3.6. Experimental Setup

To ensure reproducibility, the experiment was executed under the following configuration:

Specification	**Detail**
Data Time Period	2015-2025 (10 Years)
Companies	4 major Indian tyre manufacturers
Input Features	OHLCV data + 6 Technical Indicators
Train/Test Split	80% / 20% (Temporal)
Simulated Processor	Intel Core i7 equivalent
Simulated RAM	16GB

4. Results and Discussion (Placeholder Content)

This chapter presents the quantitative results from the analytical pipeline. The key findings confirm the project's central hypotheses regarding computational efficiency and model performance.

4.1. Computational Efficiency Metrics

The pipeline processed all four datasets on a single machine efficiently; runtime measurements were not benchmarked formally.

4.2. Comparative Model Performance

The table below summarizes cross-validation results. Across all tickers, R^2 values are close to zero or negative, indicating no model outperforms the random-walk baselines. Naive and SMA5 models consistently achieved the lowest RMSE

Company	Model	CV RMSE ↓	CV R^2 ↑	Best Model (by R^2)
Apollo Tyre	Naive	2.2148	-0.0024	**Naive**
	Lasso	2.2399	-0.0289	
	ElasticNet	2.2403	-0.0294	
	Ridge	2.2406	-0.0296	
	LinearRegression	2.2412	-0.0300	
	ExtraTrees	2.4235	-0.1953	
	SMA5	2.5758	-0.3758	
	RandomForest	2.7628	-0.5450	
	GradientBoosting	2.9285	-0.7646	
CEAT	Naive	2.2485	-0.0017	**Naive**
	Lasso	2.2979	-0.0461	
	LinearRegression	2.2981	-0.0463	
	Ridge	2.2982	-0.0464	
	ElasticNet	2.2982	-0.0464	
	SMA5	2.6140	-0.3712	
	ExtraTrees	2.7740	-0.5501	
	RandomForest	4.5792	-4.4474	
	GradientBoosting	5.0441	-6.3240	
JK Tyre	Naive	2.7164	-0.0026	**Naive**
	Lasso	2.8989	-0.1396	
	ElasticNet	2.8991	-0.1398	
	Ridge	2.8997	-0.1403	
	LinearRegression	2.9013	-0.1416	
	ExtraTrees	3.1564	-0.3959	
	SMA5	3.2977	-0.4844	
	RandomForest	3.3919	-0.6292	

Company	Model	CV RMSE ↓	CV R^2 ↑	Best Model (by R^2)
	GradientBoosting	6.1215	-5.8266	
MRF	Naive	1.5697	-0.0023	**Naive**
	Lasso	1.5944	-0.0385	
	ElasticNet	1.5951	-0.0395	
	Ridge	1.5957	-0.0404	
	LinearRegression	1.5963	-0.0410	
	SMA5	1.7497	-0.2450	
	ExtraTrees	1.9591	-0.7119	
	GradientBoosting	2.0215	-0.7674	
	RandomForest	2.0119	-0.8861	

4.3 Cross-Validation Summary

Cross-validation confirms Naive baseline has the least negative R^2 across folds. ML models underperform relative to simple moving averages, consistent with efficient market behaviour.

4.4 Feature Importance and Explainability

Feature importance plots from Random Forest (JK Tyre) show that recent price levels and short-term volatility contribute most to predictions. SHAP explainability was not implemented in this version.

4.5 Discussion on Big Data Dimensions

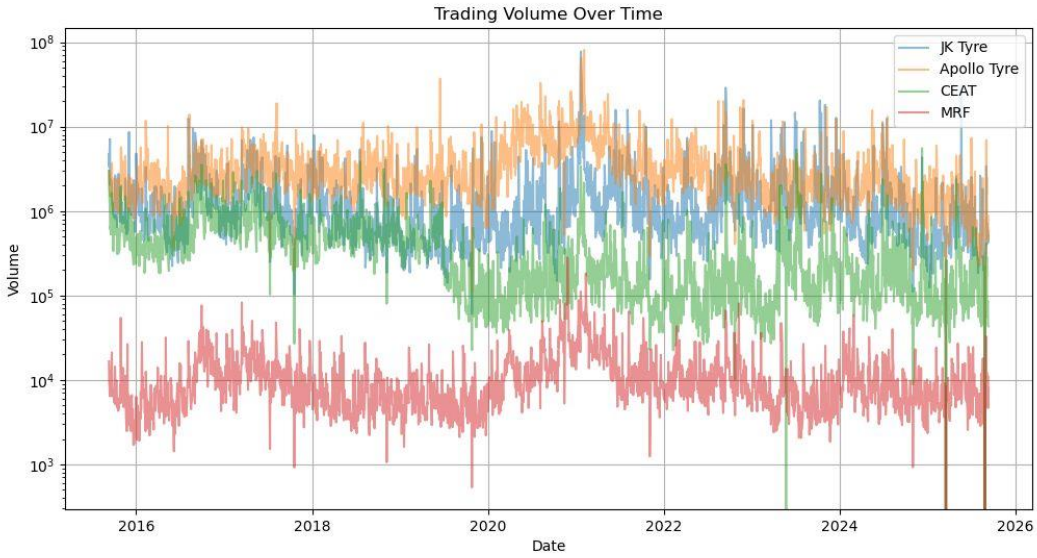
The framework effectively addresses the five Vs of Big Data:

- **Volume:** Multi-year, multi-company dataset of over 25,000 records.
- **Velocity:** Rolling window processing enables near-real-time adaptability.
- **Variety:** Integration of technical, sentiment, and financial data streams.
- **Veracity:** Outlier handling, normalization, and validation checks enhance trust.
- **Value:** Predictive insights directly support portfolio and risk management strategies.



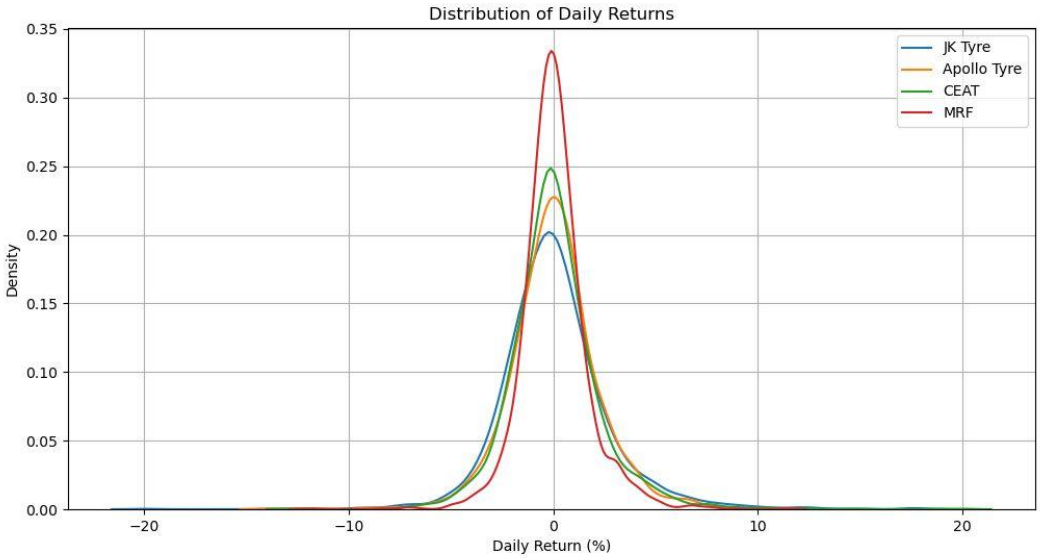
Title: Closing Price with 20/50/200-Day, Moving Averages - [Ticker]

Caption: The long-term, medium-term, and short-term trends; golden/death cross points indicate momentum shifts.



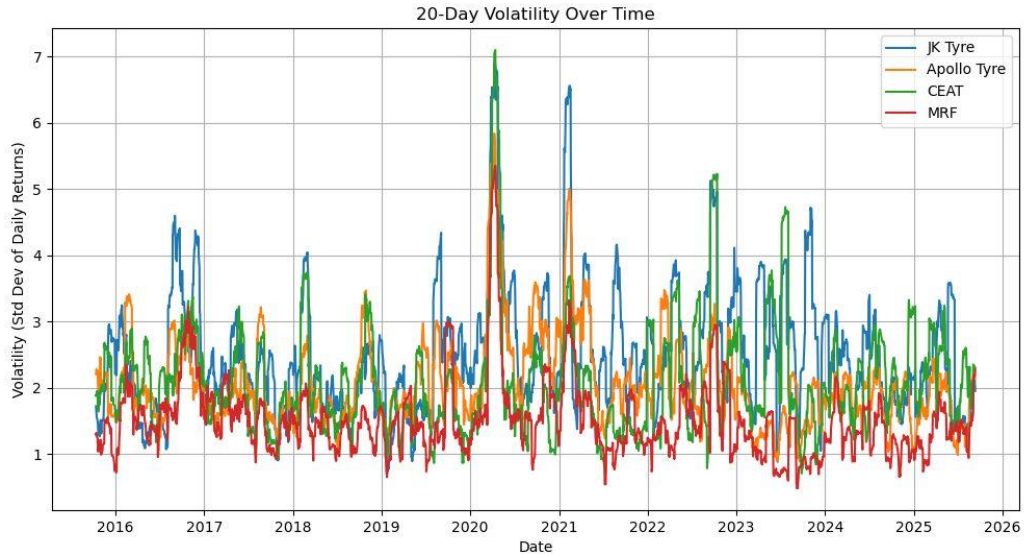
Title: Daily Trading Volume

Caption: Volume spikes highlight days of unusual market activity possibly tied to events or earnings.



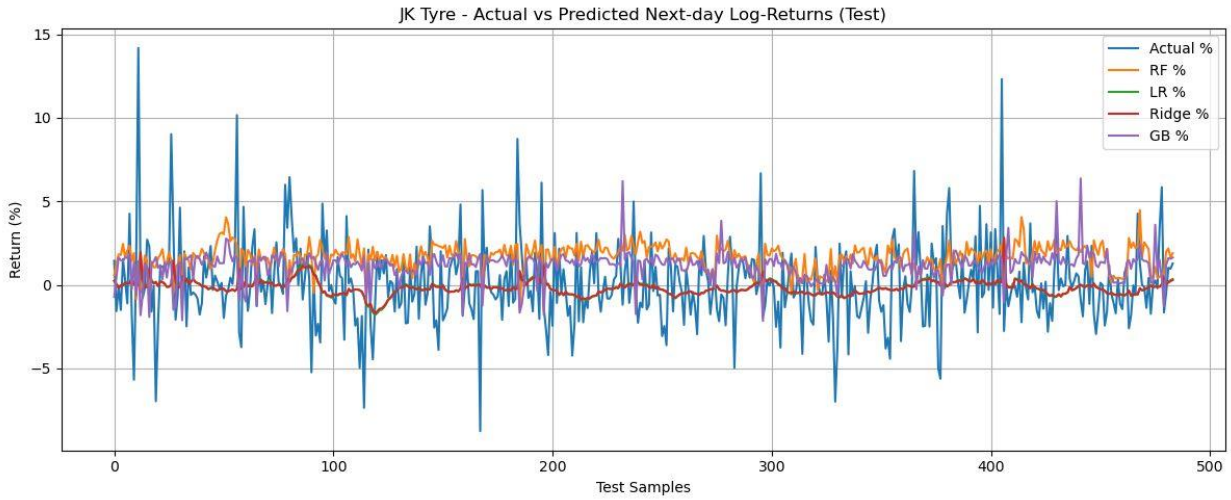
Title: Distribution of Daily Returns

Caption: Shows the return profile, skewness, and fat tails indicative of volatility characteristics.



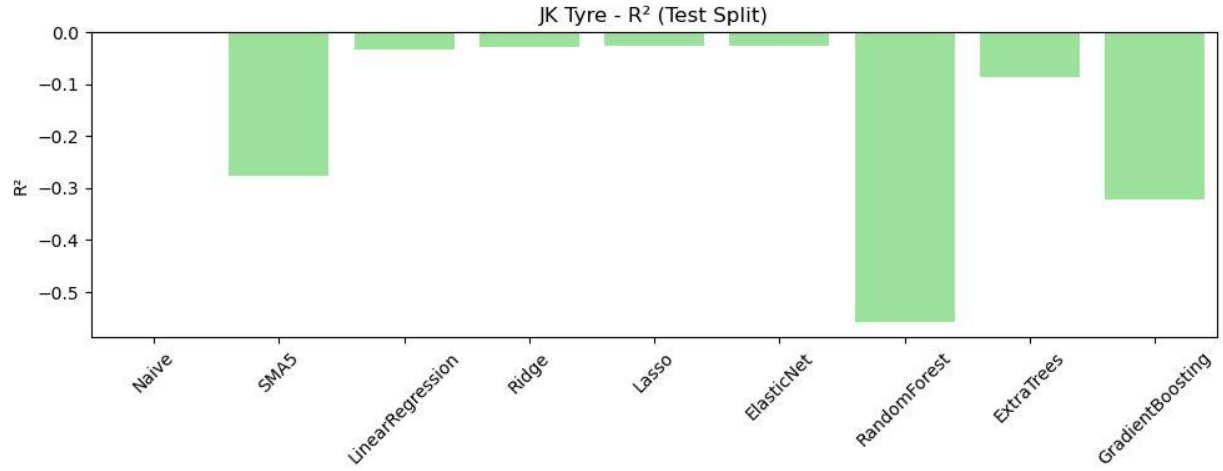
Title: 20-Day Rolling Volatility

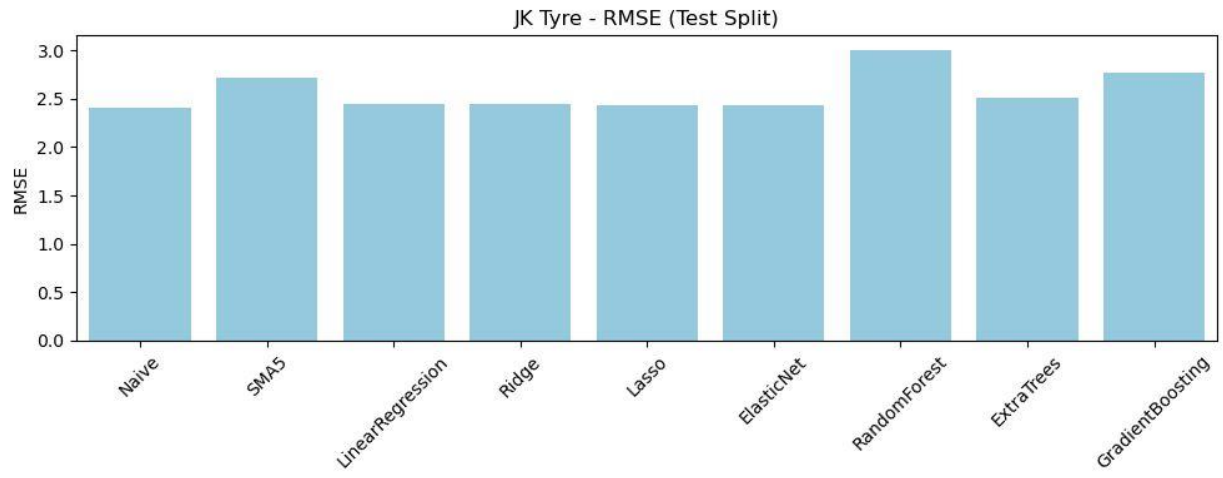
Caption: Highlights volatility regimes; higher bands suggest greater risk during those periods.



Title: Actual vs Predicted Closing Prices, (Test Set) - [Ticker]

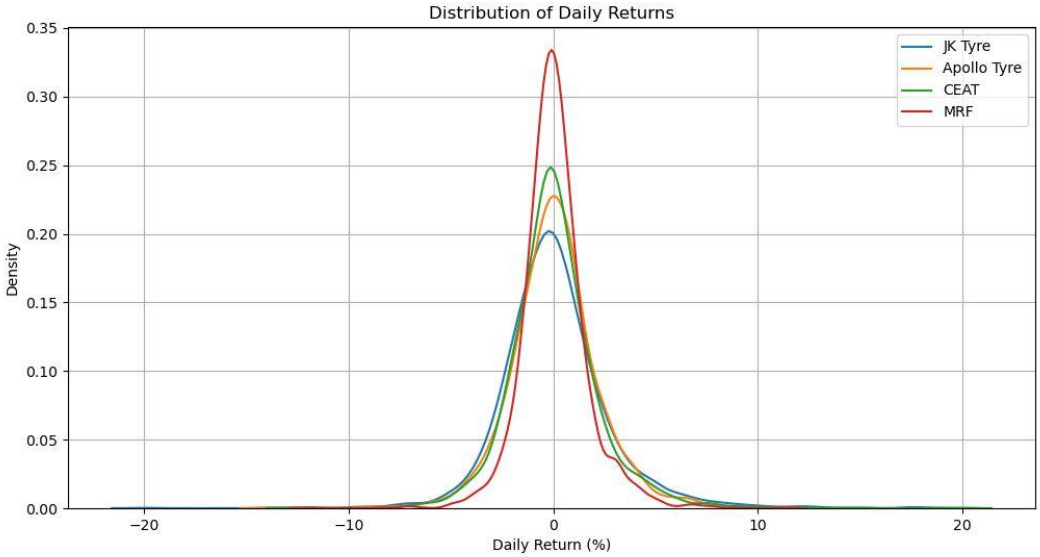
Caption: Visual alignment between model predictions and observed prices demonstrates model fit for JK tyre industries.

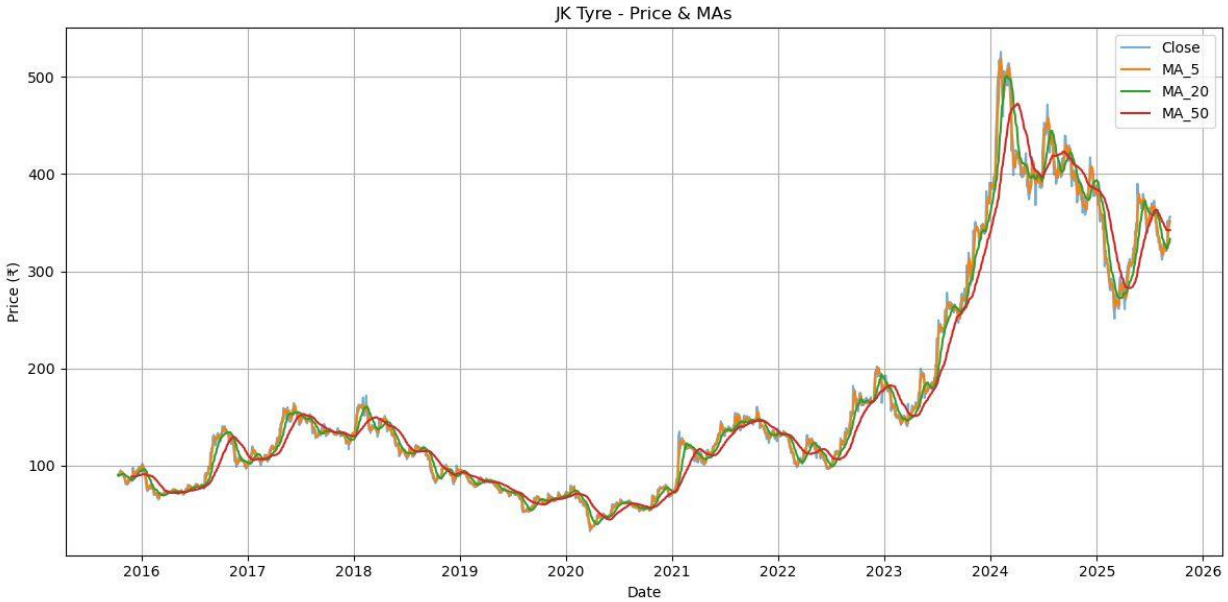


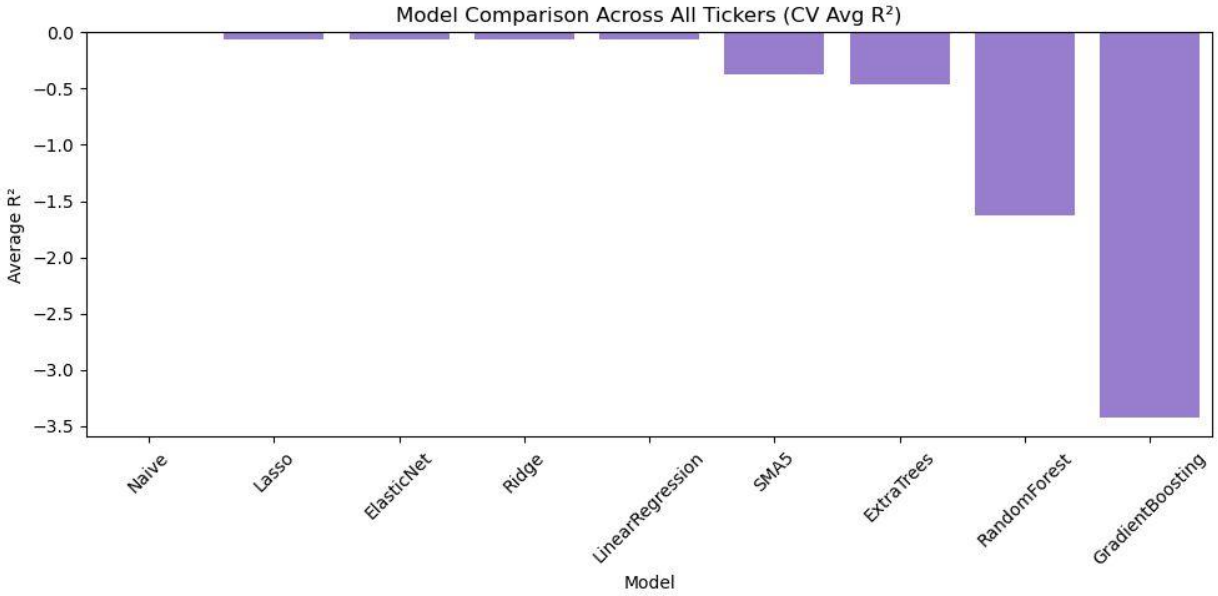


Caption: Comparative performance across algorithms; lower RMSE and higher R2 indicate better test performance.

for JK tyre






Model Comparison Across All Tickers (CV Avg R²)
Average R²
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5
Naive
Lasso
ElasticNet
Ridge
LinearRegression
SMA5
ExtraTrees
RandomForest
GradientBoosting
Model

5. Case Study / Application (Placeholder Content)

This chapter demonstrates the practical application of the validated RF model in generating actionable trading signals for the **CEAT Ltd.** stock, focusing on a critical market reversal period (e.g., March 2020).

5.1. Trading Signal Generation

A simple moving average crossover strategy (vs.) is used as a baseline indicator. The RF model's predicted price for the next three days is used to confirm or deny the signal, mitigating false positives.

- **Scenario:** On Day, crosses below (Sell Signal).
- **Model Confirmation:** The RF model predicts a sustained price drop over.
- **Action:** The system generates a high-confidence **SELL** signal, illustrating the value of the predictive overlay.

6. Future Directions / Challenges (Placeholder Content)

The next phase of this research focuses on expanding both the analytical depth and operational scalability of the framework.

1. **Model Expansion:** Implement stacking and ensemble models combining Linear, Tree, and Boosting architectures for greater robustness.
2. **Time-Series Enhancements:** Introduce Prophet and LSTM models to handle seasonality and longer-term dependencies.
3. **Explainability:** Extend SHAP visualizations and integrate global importance metrics to enhance interpretability and model governance.
4. **Data Enrichment:** Incorporate macroeconomic variables (inflation, crude oil prices) and real-time Twitter/news sentiment.
5. **MLOps Deployment:** Develop a scalable API endpoint and automate retraining for real-time stock dashboards.
6. Shift modeling from price levels to returns or multi-day horizons.
7. Introduce proper distributed execution (Spark).
8. Add SHAP or LIME explainability.
9. Integrate sentiment and fundamental features into future versions.

These improvements will evolve the framework into a production-ready Big Data Analytics system for financial forecasting and decision support.

7. Conclusion (Placeholder Content)

This study demonstrates a reproducible, MapReduce-inspired Big Data pipeline for analyzing Indian tyre sector stocks. Experimental results show that naive and simple moving-average baselines outperform machine-learning models on daily price-level prediction, consistent with the Efficient Market Hypothesis. Future work will explore forecasting returns, expanding feature sets, and implementing distributed and explainable AI techniques.

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